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Integrals depending on a parameter

2.1 Recall on integrals

In this first part, we recall the fundamental results on definite and improper integrals.

2.1.1 Definite integrals

Let a, b be two real numbers, and let the function

$$\begin{array}{ccc} f : & [a, b] & \longrightarrow \mathbb{R} \\ & t & \longmapsto f(t) \end{array}$$

Theorem 2.1.1. *If the function f is defined and continuous on the closed bounded interval $[a, b]$, then the function f is integrable on $[a, b]$, in other words, the integral*

$$I = \int_a^b f(t) dt$$

exists (or equivalently, it is finite). The integral I will be denoted indifferently as follows

$$I = \int_a^b f(t) dt \quad \text{or} \quad I = \int_{[a,b]} f(t) dt.$$

Remark 2.1.1. *It should be remembered that the two fundamental hypotheses that must really be checked in order to apply the previous theorem are that the interval $[a, b]$ is closed and bounded (we also say that it is compact), and that f is continuous at every point of $[a, b]$.*

2.1.2 Properties of an integral

Let f and g be functions from $\mathbb{R}$ to $\mathbb{R}$, piecewise continuous on the intervals under consideration. The numbers λ, μ, a, b and c are real.

- **Linearity:**

$$\int_a^b [\lambda f(x) + \mu g(x)] dx = \lambda \int_a^b f(x) dx + \mu \int_a^b g(x) dx.$$

- **Chasles' relation:**

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$

• **Order relation:**

- If $a < b$, and if $f \leq g$ on $[a, b]$, then :

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx.$$

- If f is continuous and nonnegative on $[a, b]$, we have :

$$\int_a^b f(x) dx = 0 \iff \forall x \in [a, b] \quad f(x) = 0.$$

• **Estimation of the integral:**

- If $a < b$, then

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx.$$

- If, for all $x \in [a, b]$ (with $a < b$), we have $m \leq f(x) \leq M$, then :

$$m \leq \frac{1}{b-a} \int_a^b f(x) dx \leq M.$$

The number $\frac{1}{b-a} \int_a^b f(x) dx$ is the mean value of f on $[a, b]$.

- If $a < b$, then

$$\left| \int_a^b f(x)g(x) dx \right| \leq \sup_{x \in [a, b]} |f(x)| \int_a^b |g(x)| dx.$$

In particular :

$$\left| \int_a^b f(x) dx \right| \leq |b-a| \sup_{x \in [a, b]} |f(x)|.$$

2.1.3 Locally integrable functions

Let I be an interval of $\mathbb{R}$ and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function.

Definition 2.1.1. We say that f is locally integrable on I if f is integrable on every closed bounded interval contained in I .

Remarks 2.1.1.

1. All continuous functions are locally integrable.
2. We denote by $\text{Loc}(I, \mathbb{R}) = \{f : I \rightarrow \mathbb{R} : f \text{ locally integrable}\}$.
We then have $C(I, \mathbb{R}) \subset \text{Loc}(I, \mathbb{R})$ and the inclusion is strict.

2.1.4 Improper integrals

Let $a, b \in \mathbb{R}$, $-\infty < a < b < +\infty$, and let f be a locally integrable function. There are then two types of improper integrals:

- Improper integrals of the first kind have the following form:

$$\int_{|a,b|} f(t) dt, \quad \int_{[a,b[} f(t) dt, \quad \int_{]a,b]} f(t) dt.$$

- Improper integrals of the second kind have the following form:

$$\int_{-\infty}^b f(t) dt, \quad \int_a^{+\infty} f(t) dt, \quad \int_{-\infty}^{+\infty} f(t) dt.$$

Remarks 2.1.2.

1. An improper integral of the first or second kind is not, strictly speaking, an integral, but rather the limit of an integral. For example, the improper integral $\int_a^{+\infty} f(t) dt$ is defined as follows:

$$\int_{[a,+\infty[} f(t) dt = \int_a^{+\infty} f(t) dt = \lim_{X \rightarrow +\infty} \int_a^X f(t) dt.$$

2. Since improper integrals are limits of integrals, the first problem that arises in their study is the problem of convergence. Indeed, we say for instance that the improper integral

$$\int_a^{+\infty} f(t) dt$$

converges if

$$\int_a^{+\infty} f(t) dt = \lim_{X \rightarrow +\infty} \int_a^X f(t) dt, \quad f \in \text{Loc}([a, +\infty[, \mathbb{R})$$

exists and is finite. The same is done for improper integrals of the first kind.

3. If the improper integral

$$\int_a^{+\infty} f(t) dt$$

converges, we then say that the function $f(t)$ is integrable.

4. If f is locally integrable on $]a, +\infty[$. We set

$$\int_a^{+\infty} f(t) dt = \int_a^c f(t) dt + \int_c^{+\infty} f(t) dt, \quad c \in]a, +\infty[.$$

We say that the integral $\int_a^{+\infty} f(t) dt$ converges if and only if the two integrals on the right-hand side converge.

The results stated for improper integrals on the intervals $[a, +\infty[$ and $]a, +\infty[$ adapt without difficulty to the other types of improper intervals.

$$[a, b[, \quad]a, b], \quad]a, b[, \quad]-\infty, b], \quad]-\infty, b[.$$

Example 2.1.1. *Let us study the convergence of the improper integral*

$$\int_0^{+\infty} e^{-\lambda t} dt.$$

We have

$$\begin{aligned} \int_0^{+\infty} e^{-\lambda t} dt &= \lim_{X \rightarrow +\infty} \int_0^X e^{-\lambda t} dt \\ &= \lim_{X \rightarrow +\infty} \left[\frac{e^{-\lambda t}}{-\lambda} \right]_0^X = \lim_{X \rightarrow +\infty} \left(\frac{e^{-\lambda X} - 1}{-\lambda} \right) \\ &= \begin{cases} \frac{1}{\lambda} & \text{if } \lambda > 0 \\ +\infty & \text{if } \lambda < 0 \end{cases} \end{aligned}$$

If $\lambda = 0$ then

$$\int_0^{+\infty} e^{-\lambda t} dt = \int_0^{+\infty} dt = +\infty.$$

The integral $\int_0^{+\infty} e^{-\lambda t} dt$ converges if and only if $\lambda > 0$.

Example 2.1.2. *Let us study the convergence of the improper integral*

$$\int_0^1 \frac{1}{\sqrt{1-t^2}} dt.$$

We have

$$\int_0^1 \frac{1}{\sqrt{1-t^2}} dt = \lim_{X \rightarrow 1} \int_0^X \frac{1}{\sqrt{1-t^2}} dt = \lim_{X \rightarrow 1} [\arcsin t]_0^X = \frac{\pi}{2}.$$

Remark 2.1.2. *If a function f is defined on $]a, +\infty[$ and can be extended by continuity at a , then the integral*

$$\int_a^{+\infty} f(x) dx$$

is not improper near a . Indeed, the limit defining the improper integral exists simply and the integral is ordinary near a ; the only possible impropriety may come from the behavior at infinity.

Example 2.1.3. *Consider*

$$f(t) = \frac{1 - \cos t}{t^2}, \quad t > 0.$$

We have

$$\lim_{t \rightarrow 0} \frac{1 - \cos t}{t^2} = \frac{1}{2},$$

so f can be extended by continuity at $t = 0$ by setting $f(0) = \frac{1}{2}$.

Thus, the integral

$$\int_0^{+\infty} \frac{1 - \cos t}{t^2} dt$$

*is improper only at $+\infty$; near 0, it is **proper** thanks to the extension by continuity.*

2.1.5 Convergence criteria for improper integrals

- **Comparison test:** Let f and g be locally integrable functions such that $0 \leq f \leq g$ on $[a, +\infty[$.

– If

$$\int_a^{+\infty} g(t) dt \text{ converges, then } \int_a^{+\infty} f(t) dt \text{ also converges.}$$

– If

$$\int_a^{+\infty} f(t) dt \text{ diverges, then } \int_a^{+\infty} g(t) dt \text{ also diverges.}$$

- **Riemann comparison rules:**

$$\lim_{t \rightarrow +\infty} t^\alpha f(t) = \ell \quad (\ell \neq 0 \text{ and } \ell \neq +\infty) \implies \int_1^\infty f(t) dt \text{ and } \int_1^\infty \frac{dt}{t^\alpha} \text{ have the same nature.}$$

$$\lim_{t \rightarrow +\infty} (t^\alpha f(t)) = 0, \quad \text{and} \quad \alpha > 1 \implies \int_1^\infty f(t) dt \text{ converges.}$$

$$\lim_{t \rightarrow +\infty} (t^\alpha f(t)) = +\infty, \quad \text{and} \quad \alpha \leq 1 \implies \int_1^\infty f(t) dt \text{ diverges.}$$

- **Equivalence criterion:** Let f and g be two positive functions.

– If $f(x) \sim_\infty g(x)$, then the integrals

$$\int_a^{+\infty} f(t) dt \quad \text{and} \quad \int_a^{+\infty} g(t) dt$$

have the same nature (they are either both convergent or both divergent).

– If $f(x) \sim_b g(x)$, then the integrals

$$\int_a^b f(t) dt \quad \text{and} \quad \int_a^b g(t) dt$$

have the same nature.

- **Absolute convergence:** Let f be a function locally integrable on $[a, +\infty[$. We say that f is absolutely integrable on $[a, +\infty[$ if

$$\int_a^{+\infty} |f(t)| dt \text{ converges,}$$

and we have

$$\int_a^{+\infty} |f(t)| dt \text{ converges} \implies \int_a^{+\infty} f(t) dt \text{ converges.}$$

- **Reference improper integrals**

- **Riemann integral**

$$\int_1^{+\infty} \frac{1}{t^\alpha} dt \begin{cases} \text{converges} & \text{if } \alpha > 1, \\ \text{diverges} & \text{if } \alpha \leq 1. \end{cases}$$

$$\int_0^1 \frac{1}{t^\alpha} dt \begin{cases} \text{converges} & \text{if } \alpha < 1, \\ \text{diverges} & \text{if } \alpha \geq 1. \end{cases}$$

- **Bertrand integral**

$$\int_2^{+\infty} \frac{1}{t^\alpha (\ln t)^\beta} dt \begin{cases} \text{converges} & \text{if } \alpha > 1, \\ \text{converges} & \text{if } \alpha = 1 \text{ and } \beta > 1, \\ \text{diverges} & \text{otherwise.} \end{cases}$$

- **Exponential integral**

$$\int_0^{+\infty} e^{-\lambda t} dt = \frac{1}{\lambda} \quad \text{if } \lambda > 0.$$

It diverges if $\lambda \leq 0$.

- **Singularity at a finite endpoint**

$$\int_a^b \frac{1}{(t-b)^\beta} dt \begin{cases} \text{converges} & \text{if } \beta < 1, \\ \text{diverges} & \text{if } \beta \geq 1. \end{cases}$$

- **Abel's theorem:** Let $f, g : [a, b[\rightarrow \mathbb{R}$ be two locally integrable functions.

- i) If f is monotone and $\lim_{t \rightarrow b} f(t) = 0$, and
- ii) if there exists $M > 0$ such that for all $x \in [a, b]$,

$$\left| \int_a^x g(t) dt \right| \leq M,$$

then the improper integral

$$\int_a^b f(t)g(t) dt$$

converges.

Examples 2.1.1.

1. The integral $\int_0^{+\infty} e^{-t^2} dt$

Let $I = \int_0^{+\infty} e^{-t^2} dt = I_1 + I_2$ where

- $I_1 = \int_0^1 e^{-t^2} dt$ is a proper integral hence convergent. (The function $t \mapsto e^{-t^2}$ is continuous on the closed bounded interval $[0, 1]$).
- $I_2 = \int_1^{+\infty} e^{-t^2} dt$. For all $t \in [1, +\infty[$,

$$t^2 \geq t \implies e^{-t^2} \leq e^{-t}$$

The integral $\int_1^{+\infty} e^{-t} dt$ converges so I_2 also converges by the comparison test.

We deduce that the integral I converges.

2. The integral $\int_0^1 \frac{dt}{\sqrt{t}(1-t)}$

- Near 0: $\frac{1}{\sqrt{t}(1-t)} \sim_0 \frac{1}{\sqrt{t}}$ and

$$\int_0^1 \frac{dt}{\sqrt{t}} \text{ is a convergent Riemann integral } (\alpha = \frac{1}{2} < 1),$$

hence convergent at 0.

- Near 1: $\frac{1}{\sqrt{t}(1-t)} \sim_1 \frac{1}{1-t}$ and

$$\int_0^1 \frac{dt}{1-t} = \lim_{X \rightarrow 1} [-\ln(1-t)]_0^X = -\lim_{X \rightarrow 1} \ln(1-X) = +\infty,$$

so $\int_0^1 \frac{dt}{\sqrt{t}(1-t)}$ diverges at 1.

We deduce that $\int_0^1 \frac{dt}{\sqrt{t}(1-t)}$ diverges.

3. The improper integral $\int_1^{+\infty} \frac{\sin t}{t} dt$ converges. Indeed,

$f(t) = \frac{1}{t}$ is a decreasing function on $[1, +\infty[$, and $\lim_{t \rightarrow +\infty} \frac{1}{t} = 0$.

Moreover, for all $x \in [1, +\infty[$,

$$\left| \int_1^x \sin(t) dt \right| = |\cos(1) - \cos(x)| \leq 2.$$

Thus by Abel's theorem, the improper integral $\int_1^{+\infty} \frac{\sin t}{t} dt$ converges.

2.2 Definite Integrals depending on a parameter

Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Let a and b be two real numbers, $\mathcal{D}_x \subseteq \mathbb{R}$ and consider the two-variable function

$$\begin{aligned} f : [a, b] \times \mathcal{D}_x &\longrightarrow \mathbb{K} \\ (t, x) &\longmapsto f(t, x) \end{aligned}$$

In this section we study **continuity** and **differentiability** of functions of the form

$$F(x) = \int_a^b f(t, x) dt.$$

We say that F is a **definite Integrals depending on a parameter** since its value depends on the parameter x such that for any fixed $x \in \mathcal{D}_x$, if $t \mapsto f(t, x)$ is continuous on the compact $[a, b]$ then the integral $\int_a^b f(t, x) dt$ exists.

2.2.1 Continuity of Definite Integrals depending on a parameter

Theorem 2.2.1. (*Continuity Theorem*)

Let $f : [a, b] \times \mathcal{D}_x \longrightarrow \mathbb{K}$ be continuous with respect to $(t, x) \in [a, b] \times \mathcal{D}_x$. Then the function $F : \mathcal{D}_x \longrightarrow \mathbb{K}$ defined by $F(x) = \int_a^b f(t, x) dt$ is **well-defined** and **continuous** on $\mathcal{D}_x$.

Example 2.2.1. The function F defined by

$$F(x) = \int_0^1 \frac{\sin t + e^{tx}}{t + x^2 + 1} dt$$

is continuous on $\mathbb{R}$.

2.2.2 Differentiability of Definite Integrals depending on a parameter

Theorem 2.2.2. (*Differentiability Theorem*) Let $f : [a, b] \times \mathcal{D}_x \longrightarrow \mathbb{K}$. Suppose that

1. f is continuous with respect to $(t, x) \in [a, b] \times \mathcal{D}_x$.
2. $\frac{\partial f}{\partial x}(t, x)$ exists and is continuous on $(t, x) \in [a, b] \times \mathcal{D}_x$.

Then the parameterized integral

$$F(x) = \int_a^b f(t, x) dt$$

exists and is a **differentiable** function with

$$F'(x) = \int_a^b \frac{\partial f}{\partial x}(t, x) dt$$

Example 2.2.2. Let $f : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ be such that

$$f(t, x) = \begin{cases} \frac{e^{-tx} \sin t}{t} & \text{if } t \neq 0, \\ 1 & \text{if } t = 0. \end{cases}$$

1. f is continuous on $\mathbb{R}^* \times \mathbb{R}$. Since

$$\lim_{(t,x) \rightarrow (0,x)} f(t,x) = 1,$$

f is continuous on $\mathbb{R}^2$.

2. $\frac{\partial f}{\partial x}(t,x) = -e^{-tx} \sin t$ exists and is continuous. This implies that

$$F(x) = \int_0^1 \frac{e^{-tx} \sin t}{t} dt$$

is differentiable on $\mathbb{R}$ and we have

$$F'(x) = - \int_0^1 e^{-tx} \sin t dt.$$

Theorem 2.2.3. Let $f : [a,b] \times \mathbb{R} \rightarrow \mathbb{R}$ be continuous and let $u, v : \mathbb{R} \rightarrow [a,b]$ be two functions. Then:

1. If u, v are continuous then the function

$$F(x) = \int_{u(x)}^{v(x)} f(t,x) dt$$

is continuous.

2. (a) If $\frac{\partial f}{\partial x}$ exists and is continuous.

(b) If the functions $u, v : \mathbb{R} \rightarrow [a,b]$ are differentiable then F is differentiable and we have:

$$F'(x) = \int_{u(x)}^{v(x)} \frac{\partial f}{\partial x}(t,x) dt + v'(x)f(v(x),x) - u'(x)f(u(x),x).$$

2.3 Improper Integrals depending on a parameter

Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Let a be a real number, $\mathcal{D}_x \subseteq \mathbb{R}$ and consider the two-variable function

$$\begin{aligned} f : [a, +\infty[\times \mathcal{D}_x &\rightarrow \mathbb{R} \\ (t,x) &\mapsto f(t,x) \end{aligned}$$

In this section we study the **existence**, **continuity** and **differentiability** of functions of the form

$$F(x) = \int_a^{+\infty} f(t,x) dt.$$

We say that F is a **Improper Integrals depending on a parameter**.

The results stated for improper integrals over intervals $[a, +\infty[$ adapt easily to other types of improper intervals.

2.3.1 Types of Convergence

We consider the integral:

$$F(x) = \int_a^{+\infty} f(t, x) dt.$$

1. Pointwise Convergence

Definition 2.3.1. We say the integral converges pointwise if:

$$\forall x \in \mathcal{D}_x, \quad \int_a^{+\infty} f(t, x) dt \text{ converges.}$$

In other words:

$$\forall x \in \mathcal{D}_x, \quad \lim_{X \rightarrow +\infty} \int_a^X f(t, x) dt \text{ exists and is finite.}$$

Remarks 2.3.1.

1. Pointwise convergence consists of determining the set $\mathcal{D}_x \subseteq \mathbb{R}$ for which the integral defining F converges.
2. $\mathcal{D}_x \subseteq \mathbb{R}$ is the domain of definition of F and the domain of pointwise convergence of the integral $\int_a^{+\infty} f(t, x) dt$.
3. To determine $\mathcal{D}_x$ we fix x and study the improper integral as a classical improper integral using standard convergence criteria (comparison, asymptotic equivalence, absolute convergence, etc.).

2. Uniform Convergence

Definition 2.3.2. We say the integral converges uniformly on $\mathcal{D}_x$ if:

$$\forall \varepsilon > 0, \exists c \geq a / \forall x \in \mathcal{D}_x, \left| \int_c^{+\infty} f(t, x) dt \right| \leq \varepsilon.$$

Remark 2.3.1. Generally, we reduce the study of uniform convergence of the improper integral

$$\int_a^{+\infty} f(t, x) dt$$

to the study of uniform convergence of the sequence of functions

$$F_n(x) = \int_a^{x_n} f(t, x) dt.$$

where (x_n) is a sequence in $[a, +\infty[$ such that $\lim_{n \rightarrow +\infty} x_n = +\infty$. We have

$$\int_a^{+\infty} f(t, x) dt \text{ converges uniformly} \iff \int_a^{x_n} f(t, x) dt \text{ converges uniformly.}$$

Remarks 2.3.2.

1. To show that the improper integral $F(x) = \int_a^{+\infty} f(t, x) dt$ converges uniformly, we generally proceed via normal convergence.
2. To show that the improper integral

$$F(x) = \int_a^{+\infty} f(t, x) dt$$

does not converge uniformly, we proceed as we did for series of functions, i.e. we construct a counterexample, i.e. a function

$$x = g(t)$$

such that the integral

$$\int_a^{+\infty} f(t, g(t)) dt$$

diverges.

Example 2.3.1. Study the uniform convergence of the integral $\int_0^{\infty} e^{-t^2 x} dt$, $x \in \mathbb{R}$. Let

$$f(t, x) = e^{-t^2 x}.$$

- If $x \leq 0$: the integral diverges.
- If $x > 0$: We study the convergence on the intervals $[0, a[$ and $[a, \infty[$, $a > 0$.
 - On $[0, a[$: Let $g(t) = \frac{1}{t^2}$, then

$$\int_0^{\infty} f(t, g(t)) dt = \int_0^{\infty} e^{-1} dt = \infty.$$

Thus the integral $\int_0^{\infty} e^{-t^2 x} dt$ doesn't converge uniformly on $[0, a[$.

- On $[a, \infty[$, $a > 0$: Define

$$F_n(x) = \int_a^{x_n} e^{-t^2 x} dt.$$

We choose $x_n = n$, ($\lim_{n \rightarrow \infty} x_n = \infty$), and we prove that the sequence of functions F_n converge uniformly on $[a, \infty[$.

$$F_n(x) = \int_a^n e^{-t^2 x} dt \text{ converge pointwise to } F(x) = \int_a^{\infty} e^{-t^2 x} dt.$$

For $x \geq a$, we have

$$e^{-t^2 x} \leq e^{-at^2}.$$

Hence

$$0 \leq \int_n^{+\infty} e^{-t^2 x} dt \leq \int_n^{+\infty} e^{-at^2} dt.$$

Therefore

$$0 \leq F(x) - F_n(x) = \int_n^{+\infty} e^{-t^2 x} dt \leq \int_n^{+\infty} e^{-at^2} dt.$$

Since

$$\int_n^{+\infty} e^{-at^2} dt \xrightarrow{n \rightarrow \infty} 0,$$

and this bound is independent of x , we obtain

$$\sup_{x \geq a} |F(x) - F_n(x)| \rightarrow 0.$$

Thus $F_n(x)$ converges uniformly on $[a, +\infty)$, which implies that

$$\int_a^{+\infty} e^{-t^2 x} dt$$

converges uniformly on $[a, +\infty)$. □

Theorem 2.3.1. (Abel's Rule for Uniform Convergence)

Let $f : [a, b[\times I \rightarrow \mathbb{R}_+$ be a positive function and let $g : [a, b[\times I \rightarrow \mathbb{R}$ be locally integrable with respect to $t \in [a, b[$. Suppose:

(i) For all $x \in I$, the function $t \mapsto f(t, x)$, $t \in [a, b[$, is decreasing.

(ii) $\lim_{t \rightarrow b} \left(\sup_{x \in I} f(t, x) \right) = 0.$

(iii) There exists $M > 0$ such that for all $u \in [a, b[$ we have $\left| \int_a^u g(t, x) dt \right| \leq M.$

Then

$$\int_a^b f(t, x) g(t, x) dt$$

converges uniformly.

Example 2.3.2. Consider the integral $F(x) = \int_0^\infty \frac{e^{-tx}}{x+t} dt$, for $x \geq 1$.

Let $f(t, x) = \frac{1}{x+t}$ and $g(t, x) = e^{-tx}$. These functions are continuous.

We have

- $0 \leq f(t, x) \leq \frac{1}{1+t} \xrightarrow{t \rightarrow 0} 0.$
- $\forall x \in [1, \infty[,$ the function $t \mapsto f(t, x), t \in [0, \infty[$ is decreasing.
- $\forall u \in [0, \infty[$ we have

$$\begin{aligned} \left| \int_0^u g(t, x) dt \right| &= \left| \left(-\frac{e^{-tx}}{x} \right)_0^u \right| \\ &= \left| \frac{1 - e^{-ux}}{x} \right| \leq 2. \end{aligned}$$

The conditions of Abel's rule are satisfied then the integral converge uniformly on $[1, \infty[$.

Corollary 2.3.1. Let $f : \mathbb{R} \rightarrow \mathbb{R}_+$ be a positive decreasing function satisfying $\lim_{x \rightarrow +\infty} f(x) = 0.$

Then

$$F(x) = \int_0^{+\infty} f(t) e^{itx} dt$$

converges uniformly on $[a, +\infty[$ for all $a > 0.$

3. Normal Convergence

Definition 2.3.3. Suppose that for all $x \in \mathcal{D}_x$, the map $t \mapsto f(t, x)$ from $[a, +\infty[$ to $\mathbb{R}$ is locally integrable. If there exists a locally integrable map

$$g : [a, +\infty[\rightarrow \mathbb{R}_+$$

satisfying:

$$(i) \int_a^{+\infty} g(t) dt \text{ converges;}$$

$$(ii) \forall (t, x) \in [a, +\infty[\times \mathcal{D}_x, \quad |f(t, x)| \leq g(t);$$

then the integral $\int_a^{+\infty} f(t, x) dt$ is said to be normally convergent on $[a, +\infty[$.

Remark 2.3.2. We have the following implications:

$$\text{Normal Convergence} \implies \text{Uniform Convergence} \implies \text{Pointwise Convergence.}$$

2.3.2 Continuity of Improper Integrals depending on a parameter

Theorem 2.3.2. (Continuity Theorem)

Let $f : [a, +\infty[\times \mathcal{D}_x \rightarrow \mathbb{R}$ be continuous with respect to $(t, x) \in [a, +\infty[\times \mathcal{D}_x$. Suppose $F(x) = \int_a^{+\infty} f(t, x) dt$ converges uniformly then F is **well-defined** and **continuous** on $\mathcal{D}_x$.

Example 2.3.3. Show that $F(x) = \int_0^{+\infty} \sin(xt)e^{-t^2} dt$ is well-defined and continuous for $x \in \mathbb{R}$.

- For fixed x , $t \mapsto \sin(xt)e^{-t^2}$ is continuous and

$$\lim_{t \rightarrow +\infty} t^2 \sin(xt)e^{-t^2} = 0,$$

so $\int_0^{+\infty} \sin(xt)e^{-t^2} dt$ converges pointwise hence F is well-defined.

- Let $f(t, x) = \sin(xt)e^{-t^2}$.
 - f is continuous with respect to $(t, x) \in [0, +\infty[\times \mathbb{R}$.
 - For each $(t, x) \in [0, +\infty[\times \mathbb{R}$: $|f(t, x)| \leq e^{-t^2} = g(t)$. Moreover g is continuous and integrable on $[0, \infty[$ then the integral converge uniformly on $\mathbb{R}$.

By the continuity theorem for parameterized integrals, F is continuous on $\mathbb{R}$.

2.3.3 Differentiability of Improper Integrals depending on a parameter

Theorem 2.3.3. (*Differentiability Theorem*) Let $f : [a, +\infty[\times \mathcal{D}_x \longrightarrow \mathbb{K}$. Suppose that

1. f is continuous with respect to $(t, x) \in [a, +\infty[\times \mathcal{D}_x$.
2. $\forall x \in \mathcal{D}_x$, $\int_a^{+\infty} f(t, x) dt$ converges pointwise
3. $\frac{\partial f}{\partial x}(t, x)$ exists and is continuous on $(t, x) \in [a, +\infty[\times \mathcal{D}_x$.
4. $\int_a^{+\infty} \frac{\partial f}{\partial x}(t, x) dt$ converges uniformly on $\mathcal{D}_x$

Then $F(x) = \int_a^{+\infty} f(t, x) dt$ is **differentiable** on $[a, +\infty[$ and

$$F'(x) = \int_a^{+\infty} \frac{\partial f}{\partial x}(t, x) dt$$

Example 2.3.4. Show that $F(x) = \int_0^{+\infty} \sin(xt)e^{-t^2} dt$ is differentiable on $\mathbb{R}$.

- the function $f(t, x) = \sin(xt)e^{-t^2}$ is continuous with respect to $(t, x) \in [0, +\infty[\times \mathbb{R}$.
- We have already shown that the integral converges uniformly on $\mathbb{R}$ so it converges pointwise on $\mathbb{R}$.
- $\frac{\partial f}{\partial x}(t, x) = t \cos(xt)e^{-t^2}$ exist and it is continuous with respect to $(t, x) \in [0, +\infty[\times \mathbb{R}$.
- We have $\left| \frac{\partial f}{\partial x}(t, x) \right| \leq te^{-t^2} = g(t)$. We show that g is integrable on $[0, \infty[$.

$$\int_0^{\infty} te^{-t^2} dt = \int_0^1 te^{-t^2} dt + \int_1^{\infty} te^{-t^2} dt$$

The first integral is a definite integral so it converges, the second converges also because

$$\lim_{t \rightarrow \infty} t^2 g(t) = 0.$$

So g is integrable on $[0, \infty[$, then the integral $\int_0^{\infty} \frac{\partial f}{\partial x}(t, x) dt$ converge uniformly on $\mathbb{R}$.

From the differentiability theorem of improper integrals depending on a parameter we conclude that F is differentiable on $\mathbb{R}$ and

$$F'(x) = \int_0^{\infty} t \cos(xt)e^{-t^2} dt$$